Sentiment Analysis of the nationalization vs. privatization debate in legislative elections manifestos

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1 Introduction

The data transcribed from the Archelec corpus enables us to study electoral manifestos for the 1981, 1988 and 1993 legislative elections in France. Politics in France in the 1980s are characterized by a debate about the nationalization and privatization of companies. At the beginning of the decade, the National Assembly and the government are left-wing and nationalize industrial companies such as Saint-Gobain and banks such as the Société Générale. But at the end of the decade, many companies (including the Société Générale) are in turn privatized, in a context of cohabitation between a left-wing president and a right-wing National Assembly. In the beginning of the 1990s, the president is also right-wing, and privatizations go even further. Therefore, deputees' opinions about whether to nationalize or privatize sectors of the industry can be seen as an interesting point in a legislative manifesto.

In this project, I use the Archelec data in order to try and answer the following question: how is the nationalization vs. privatization debate mobilized in legislative election manifestos? I will focus on the evolution of this debate over time and its grappling by different party and opinion blocks, notably studying whether candidates uphold their side of the debate or criticize the opposite side in their manifestos.

2 Pre-processing of the data used

I began by pre-processing the Archelec transcriptions for the legislative manifestos of the years 1981, 1988 and 1993. When adding the metadata of the candidates to the manifesto transcriptions, I chose to include their names, party affiliation and locality. Then I added a first filter to select the manifestos that use one of the main terms studied: nationalization, etatization and privatization. I also included variations considering conjugation of each verb and accents on the letters. Lemmatizing every single manifesto seemed too costly and not efficient enough in this situation. I ended up having a dataset containing 1030 manifestos.

In order to correctly pre-process the party affiliation of the candidates, I created 6 main blocks : far-left, left-wing, center, right-wing, far-right and unaffiliated (regionalist parties, no supporting party). In order to sort candidates into their blocks, I used keywords in their affiliation party names: for example, having the word "marxiste" in the affiliation list puts a candidate in the far-left block. This enabled me to sort most of the candidates, and I hand-sorted the few exceptions. I also tagged candidates who were supported by some of the main parties at the time: the *Parti Communiste Français (PCF)*,

the *Parti Socialiste (PS)*, the *Rassemblement Pour la République (RPR)* and the *Union pour la Démocratie Française (UDF)*. These parties support a significant share of the candidates.

After adding the political affiliation of each manifesto, I created a dataset of phrases that directly tackled the themes of nationalization and privatization. I reused the previous keyword list in order to do so. I tagged specifically whether the sentences concerned nationalization, privatization or both. This final dataset contains 1211 phrases.

Finally, I randomly sampled 10% of the dataset (i.e. 121 phrases) in order to annotate a test set for the sentiment analysis classification that I will perform later on. I rated the sentiment of each phrase on its topic by the following gradings: 1 for a negative sentiment, 2 for a neutral sentiment and 3 for a positive sentiment.

3 Descriptive analysis

Before diving into the sentiment analysis of the statements about nationalization and privatization, we can draw descriptive statistics about the manifesto corpus. As said before, I am studying 1030 manifestos, 465 of which are from 1981, 69 from 1988 and 496 from 1993. We can see that over the years, the debate fades from the public opinion but then regains importance. Besides, 445 manifestos of the corpus are from far-left candidates, which means that the debate is predominantly carried by this block. Finally, 525 manifestos mention nationalization and 539 of them mention privatization, with 34 manifestos mentioning both themes.

Afterwards, we can study how the focus of the debate changes from one election to another. Figure 1 shows the number of manifestos mentioning one of the themes for each year. We can see that as time goes on, the argument becomes less about whether or not to nationalize companies and more about whether or not to privatize them. We can also conduct this analysis for each political block. Figure 2 shows that as time goes on, all blocks talk less about nationalization, with a small rebound in 1993. They talk more about privatization, especially the far-left block.

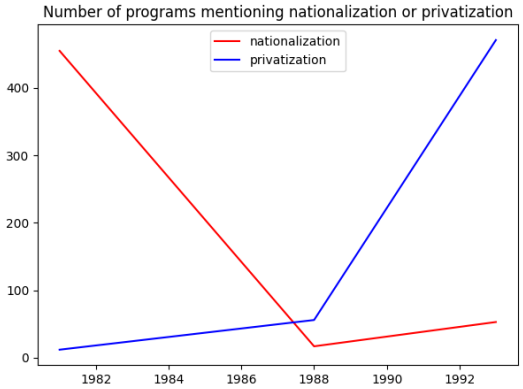

Figure 1: Evolution of the mobilization of the nationalization vs. privatization debate.

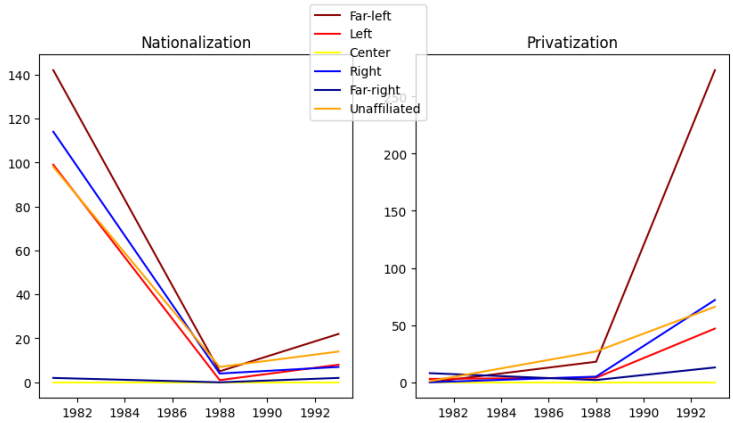

Figure 2: Evolution of the use of nationalization and privatization terms by political blocks.

From this descriptive analysis, we can draw the following hypothesis: in the beginning, the left wing supports nationalizations and the right wing criticizes them. But as time goes on, the debate becomes mostly far-left candidates criticizing privatizations, while right-wing candidates vaguely support them. We will see whether or not this hypothesis is true using sentiment analysis.

Besides this hypothesis about the sentiment of the parties on this debate, we can analyze it in a geographic perspective. Figure 3 shows a map of the number of phrases (not manifestos) about the debate for each department. We can see that the debate is prevalent in the Parisian region as well as in the north of France. This could be because this used to be a very industrialized region. The debate is also important in the Rhône department, maybe because it is the department of the second biggest city in France, Lyon.

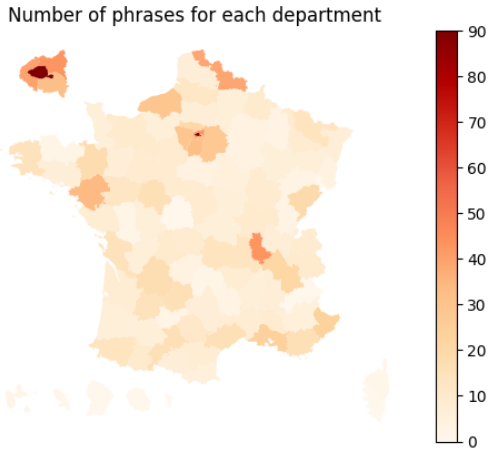

Figure 3: Mobilization of the nationalization vs. privatization debate in French localities.

Finally, we can move on from the descriptive analysis and perform some topic modeling on the phrases concerning nationalization and privatization. This could help us see which words the debate is associated with, notably which kinds of structures are expected to be nationalized or privatized. Therefore I used the Latent Dirichlet Allocation in order to retrieve 10 topics. In terms of stop words, I took out from the corpus both the classic ones such as articles and keywords about nationalization

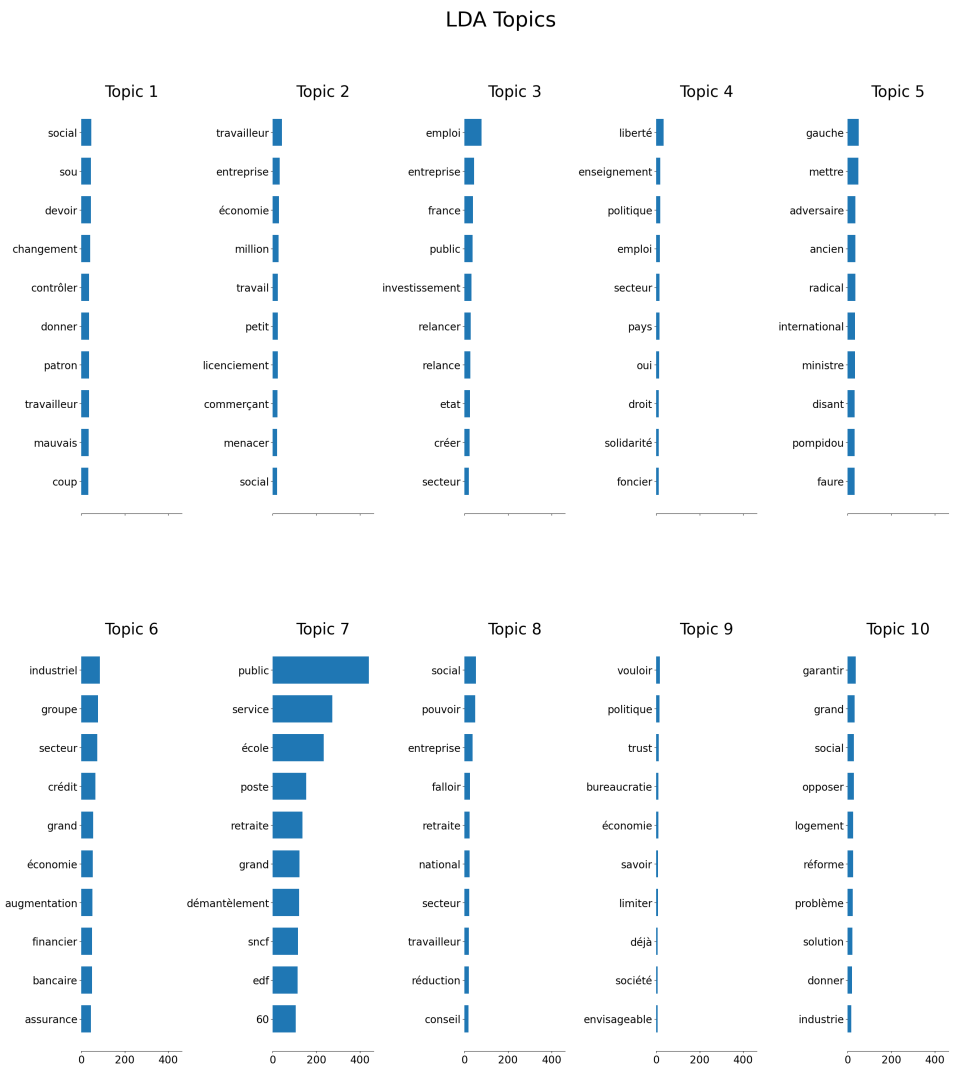

Figure 4: Results of the topic modeling on phrases concerning nationalization or privatization.

and privatization. This prevents the topics from being only about those keywords which did not interest me.

The results of the topic modeling are presented in figure 4. Some of the topics seem vague and not suited for an analysis. Topics 6 and 7 seem to be about the structures that could be nationalized or privatized: schools, banks, insurance companies, the postal office, train company (SNCF) and energy company (EDF). Topics 1 and 2 seem to be about the opposition between workers and employers. Topic 3 seems to be about the benefits of one of the positions, and topic 9 about its drawbacks. Finally, topic 5 seems to be about the opposition between parties or with the current government. Most interestingly, the topic modeling reveals that some of the most frequent words are very loaded in sentiment, whether it is to support or criticize a position.

The fact that some very positive and negative terms appear in the topic modeling makes the idea of conducting a sentiment analysis of the phrases all the more relevant. From now on, the objective will be to see how much candidates support their own position in the debate and how much they criticize their opponents'.

4 Literature review

I then searched the literature in order to gain insight on how to tackle the sentiment analysis of the corpus. The literature highlights how phenomena such as the development of online reviews and social media have made automated sentiment analysis models easier to use and more powerful. It also highlights the difficulty of performing sentiment analysis in languages other than English, using models whose training data is mostly in English.

[1] compares a lexicon-based approach with BERT for sentiment analysis of 600 customer reviews in Italian. On the one hand, it uses two lexicons: the hand-built lexicon SentIta and the automatically built Sentix. A lexicon is a series of words annotated with the polarity (positive or negative) and intensity of their sentiment. On the other end, the authors use BERT, taking advantage of its ability of bidirectional context analysis, which helps specialize its pre-training knowledge to the specific task of sentiment analysis. After using all models to classify the test set, the authors use the usual metrics (accuracy, F1-score, precision and recall) to evaluate their performances. Even though the results show that lexicon-based models are better at sentiment analysis when the datasets are small and the computational resources limited (which seemed to be the case in this article), the authors state that BERT has more potential because of its ability to grasp a complex context, with sometimes opposite sentiments stated in the same phrase, or neutral statements that are difficult to classify.

[3] compares three different versions of BERT and evaluates their ability to classify the sentiment of Twitter data. The three models are fine-tuned bert-base-multilingual-uncased, RoBERTuito (a model fine-tuned on Spanish text) and RuBERT (a model fine-tuned on Russian text). The three models are tasked with classifying 1,5 million of tweets as positive or negative. Results show that the multilingual and Spanish version of BERT perform best, the latter highlighting the importance of fine-tuning a model before testing it. Overall, this experiment demonstrates the viability of pretrained BERT models for sentiment analysis.

Finally, [2] takes a different direction and experiments with the ability of Large Language Models (LLMs) to perform sentiment analysis. It first states that LLMs are able to classify tweets as positive or negative, as [3] did with pre-trained BERT models. Then the author prompts GPT-4 with classifying a statement of the Federal Open Market Committee, the institution that sets US interest rates, as "hawkish" or "dovish". It concludes that LLM are useful and time-saving at extracting sentiment from text, even though they require some human oversight.

From the literature review, I took the ideas of comparing various fine-tuned versions of BERT with a LLM-based classifier. I chose to evaluate the performances of each model at classifying the annotated test set. Then I could use the best model to perform my sentiment analysis of the nationalization vs. privatization debate.

5 Presentation of the methods used

After reading the literature, I decided to compare three pre-trained language models :

- **Baseline BERT for sentiment analysis.** This is a BERT-base-multilingual-uncased model finetuned for sentiment analysis on product reviews in six languages: English, Dutch, German, French, Spanish, and Italian. It predicts the sentiment of the review as a number of stars (between 1 and 5). I mapped the ratings in order to keep them between 1 and 3, merging the 1-2 and 4-5 ratings. It is said to have a 0.94 off-by-1 accuracy on French reviews. The model is available at *this link*.

- **A version of BERT fine-tuned for French Sentiment analysis.** This model is a fine-tuned version of CamemBERT-base on the French subset of the *tweet_sentiment_multilingual* dataset, augmented with additional curated sentiment data from various sources. It uses three classes, the same that I used for the annotation of the test set: 1 for negative sentiment, 2 for neutral sentiment, 3 for positive sentiment. It is said to have a 0.77 accuracy on its evaluation set. The model is available at *this link*.

- **The Large Language Model Llama3.3.** I used the latest version of this LLM available on the API provided by ENSAE. It is a model of size 70B, used for complex reasoning and the analysis of nuanced phrases. I prompted it with the lines of the dataset, and the following instructions [1]:

  > You are a CONCISE political science researcher.
  > Analyze the following sentence and give its sentiment on privatization or nationalization.
  > 1 = NEGATIVE, 2 = NEUTRAL, 3 = POSITIVE.
  > Output : <integer 1-3>
  > Only provide a rating with no explanation whatsoever.

I tested each model on the annotated test set, and then compared their results with my own classification. The performances of each model are available in Table 1, while their confusion matrix are available in Figure 5.

Class	Precision			Recall			F1-Score			Support
	BERT 1	BERT 2	Llama	BERT 1	BERT 2	Llama	BERT 1	BERT 2	Llama	
1	0.82	0.88	0.95	0.57	0.84	0.94	0.67	0.86	0.94	81
2	0.31	0.00	0.44	0.62	0.00	0.50	0.42	0.00	0.47	8
3	0.45	0.64	0.84	0.69	0.78	0.84	0.54	0.70	0.84	32

Table 1: Performances of the language models at Sentiment Analysis.

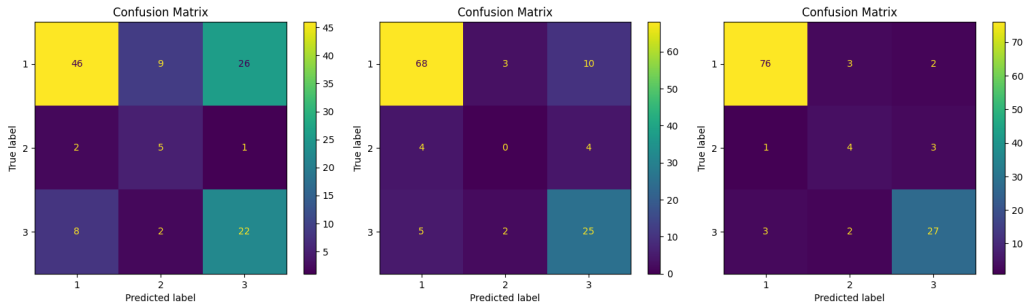

Figure 5: Confusion matrixes of the Baseline BERT model (left), the fine-tuned BERT model (center) and the Llama model (right).

It is clear that all models have trouble classifying neutral statements, but it is not a problem as they are a minority in the test set and probably the entire corpus as well. The multilingual version of BERT is less efficient than the fine-tuned one, which highlights the importance of fine-tuning. Overall, one can see that the LLM has the best classification performances. I did try other pre-trained BERT models, but they all had roughly the same performances as the second model presented above, therefore I

[1] This version is translated from French: the original prompt is available in the code.

chose Llama as my classifier. I applied this model to all the phrases in my corpus in order to be able to analyze their sentiment.

6 Results

Now that all the phrases have been classified in terms of sentiment towards nationalization or privatization, I could explore the corpus in a new perspective. I began by creating a "sentiment score". I remapped the ratings given by the LLM: a phrase is classified as -1 if it is negative, as 0 if it is neutral, and as +1 if it is positive. Therefore, by summing all scores of a certain theme, for a certain block or party, I can conclude on its general sentiment.

Let us begin by studying the sentiment of each political block, shown in figure 6. This figure takes into account all three election years. We can see that far-left parties are intensely opposed to privatization. Otherwise, far-left and left-wing parties are in support of nationalization and opposed to privatization, while right-wing parties are opposed to nationalization and in support of privatization. Center parties seem to have no opinion on the debate. Unaffiliated candidates seem too heterogeneous to have a clear stance on the debate.

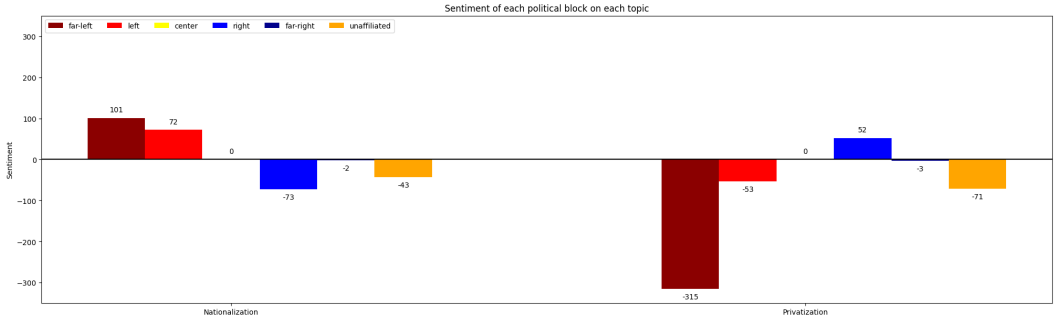

Figure 6: Sentiment score of each political block on nationalization and privatization.

We can also study the sentiment of the four main parties: *Parti Communiste Français*, *Parti Socialiste*, *Rassemblement Pour la République* and *Union pour la Démocratie Française*. Their sentiment scores are shown in figure 7. Again, the PCF is strongly opposed to privatization. Overall, the sentiment scores for the political blocks seem to follow those of the main parties that they contain, even though they also contain multiple smaller parties.

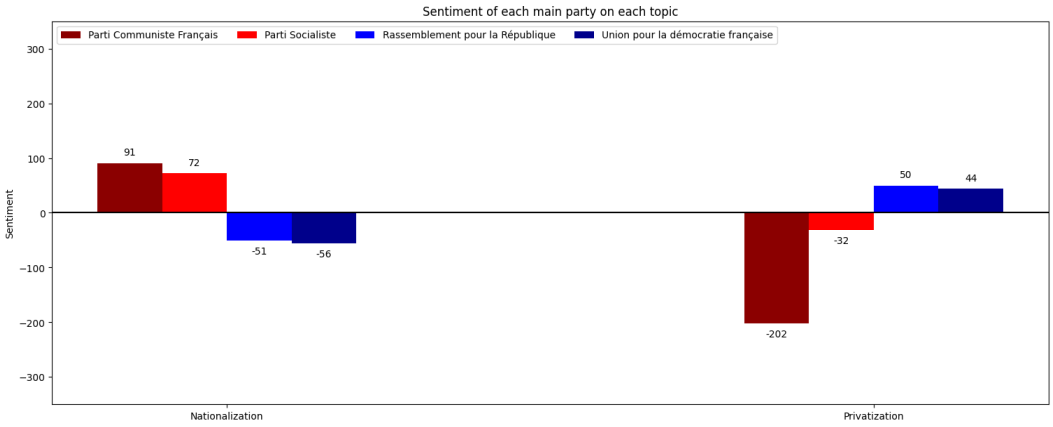

Figure 7: Sentiment scores of the main four parties.

Finally, we can study the temporal evolution of the sentiment of each political block, as represented in figure 8. When it comes to nationalization, all blocks are first very polarized, and then have a less intense stance as the years go on. When it comes to privatization, they are first not polarized, and develop an opinion in the last election. The far-left block becomes especially negatively polarized.

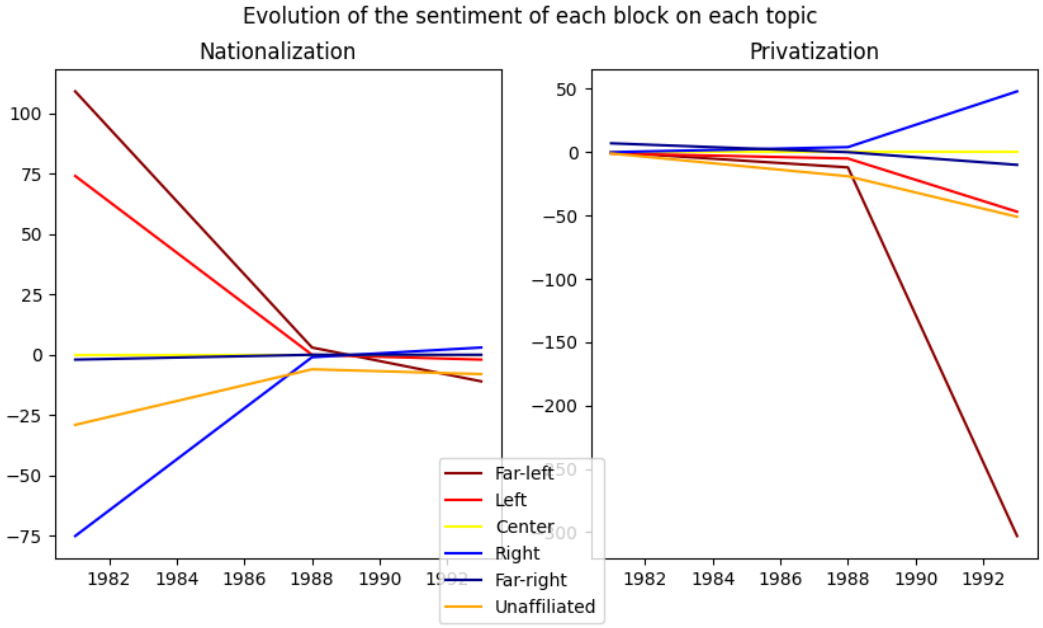

Figure 8: Evolution of the sentiment scores for each block.

7 Conclusion

The use of sentiment analysis methods have enabled us to better understand the nationalization vs. privatization debate in the 1980s and 1990s legislative elections. A first descriptive analysis and topic modelling phase helped to better understand which regions of the country and which sectors of the

economy it concerned most. After testing three different pre-trained models, I used a LLM in order to classify the sentiment of phrases including a keyword related to this debate. This classification reveals that the focus of the debate has changed from nationalization to privatization as the years went on. It also showed that this debate was driven by the far-left block, as I hypothetized in the descriptive analysis phase. In order to continue the analysis and add a new perspective, we could compare this classification with the results of each election, in order to see if a candidate's stance on the debate influences his or her chances of getting elected.

References

[1] Rosario Catelli, Serena Pelosi, and Massimo Esposito. "Lexicon-Based vs. Bert-Based Sentiment Analysis: A Comparative Study in Italian". In: *Electronics* 11.3 (2022), p. 374. ISSN: 2079-9292. DOI: 10.3390/electronics11030374.

[2] Anton Korinek. "Generative AI for Economic Research: Use Cases and Implications for Economists". In: *Journal of Economic Literature* 61.4 (2023), pp. 1281–1317. DOI: 10.1257/jel.20231736. URL: https://www.aeaweb.org/articles?id=10.1257/jel.20231736.

[3] Arghya Sahoo et al. *Comparative Analysis of BERT Models for Sentiment Analysis on Twitter Data*. Dec. 2023. DOI: 10.1109/ICSCC59169.2023.10335061.